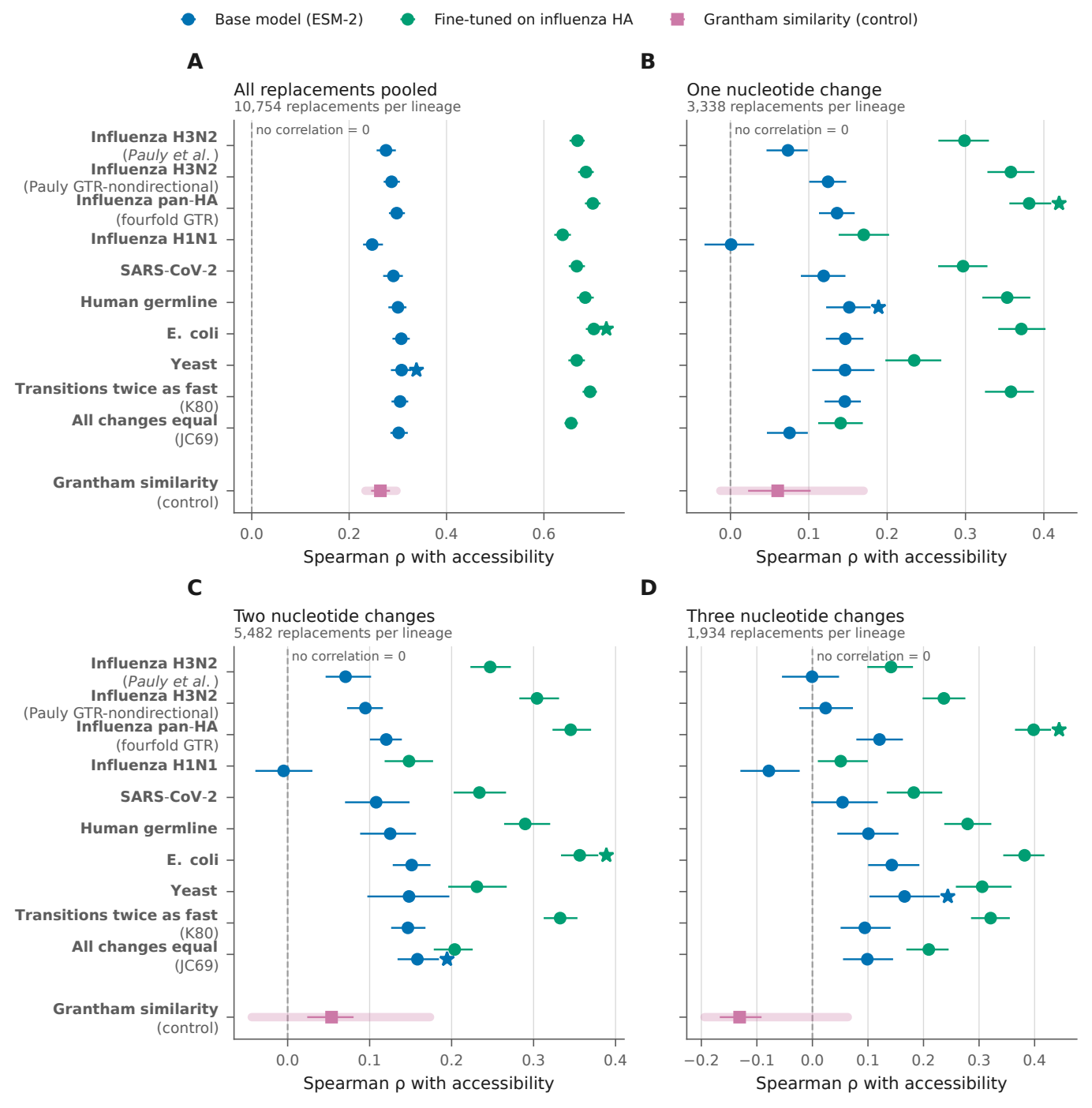
Base model (ESM-2)
Fine-tuned on influenza HA
Grantham similarity (control)
A
All replacements pooled
10,754 replacements per lineage
no correlation = 0
Influenza H3N2 (Pauly et al.)
Influenza H3N2 (Pauly GTR-nondirectional)
Influenza pan-HA (fourfold GTR)
Influenza H1N1
SARS-CoV-2
Human germline
E. coli
Yeast
Transitions twice as fast (K80)
All changes equal (JC69)
Grantham similarity (control)
0.0
0.2
0.4
0.6
Spearman ρ with accessibility
B
One nucleotide change
3,338 replacements per lineage
no correlation = 0
0.0
0.1
0.2
0.3
0.4
Spearman ρ with accessibility
C
Two nucleotide changes
5,482 replacements per lineage
no correlation = 0
Influenza H3N2 (Pauly et al.)
Influenza H3N2 (Pauly GTR-nondirectional)
Influenza pan-HA (fourfold GTR)
Influenza H1N1
SARS-CoV-2
Human germline
E. coli
Yeast
Transitions twice as fast (K80)
All changes equal (JC69)
Grantham similarity (control)
0.0
0.1
0.2
0.3
0.4
Spearman ρ with accessibility
D
Three nucleotide changes
1,934 replacements per lineage
no correlation = 0
−0.2
−0.1
0.0
0.1
0.2
0.3
0.4
Spearman ρ with accessibility